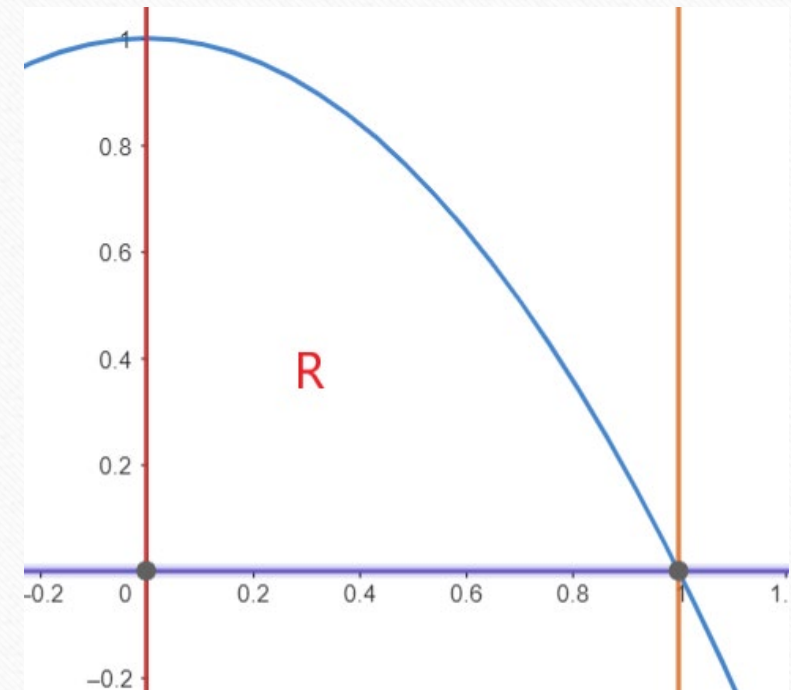


5-1 Area and Estimating with Finite Sums

師大工教一

Q: How can we find the area A of the region R enclosed by $y = 1 - x^2$, x -axis, $x = 0$, $x = 1$?



Method 1: **Upper Sums** (see Figure 5.2)

$$A \approx 1 \cdot \frac{1}{2} + \frac{3}{4} \cdot \frac{1}{2} = \frac{7}{8} = 0.875$$

$$A \approx 1 \cdot \frac{1}{4} + \frac{15}{16} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{4} + \frac{7}{16} \cdot \frac{1}{4} = \frac{25}{32} = 0.78125$$

Method 2: **Lower Sums**(see Figure 5.3(a))

$$A \approx \frac{15}{16} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{4} + \frac{7}{16} \cdot \frac{1}{4} + 0 \cdot \frac{1}{4} = \frac{17}{32} = 0.53125$$

$$0.53125 < A < 0.78125$$

Method 3: **Midpoint rule**(see Figure 5.3(b))

$$A \approx \frac{63}{64} \cdot \frac{1}{4} + \frac{55}{64} \cdot \frac{1}{4} + \frac{39}{64} \cdot \frac{1}{4} + \frac{15}{64} \cdot \frac{1}{4} = \frac{172}{64} \cdot \frac{1}{4} = 0.671875$$

HW5-1

- HW: 2,3

5-2 Sigma Notation and Limits of Finite Sums

師大工教一

Limits of Finite Sums

Ex5(p321) Find the limiting value of lower sum approximations to the area of the region R below the graph $y = 1 - x^2$ and above the interval $[0,1]$ on the x -axis using equal-width rectangles whose widths approach zero and whose number approaches infinity.

$$[0,1] \rightarrow \left[0, \frac{1}{n}\right], \left[\frac{1}{n}, \frac{2}{n}\right], \dots, \left[\frac{n-1}{n}, \frac{1}{n}\right]$$

$$f\left(\frac{1}{n}\right) \cdot \frac{1}{n} + f\left(\frac{2}{n}\right) \cdot \frac{1}{n} + \dots + f\left(\frac{n}{n}\right) \cdot \frac{1}{n} = \sum_{k=1}^n f\left(\frac{k}{n}\right) \cdot \frac{1}{n}$$

$\vdots$

$$= 1 - \frac{2n^3 + 3n^2 + n}{6n^3}$$

Taking the limit...

Riemann Sums

Dividing $[a, b]$ into n subintervals with endpoints $a = x_0 < x_1 < x_2 < \cdots < x_n = b$

$P = \{x_0, x_1, x_2, \dots, x_{n-1}, x_n\}$ is called a **partition** of $[a, b]$.

$$\mathcal{P} = \{x_0, x_1, x_2, x_n\}$$

The k -th subinterval is $[x_{k-1}, x_k]$, and the width of k -th subinterval is

$\Delta x_k = x_k - x_{k-1}$. In $[x_{k-1}, x_k]$, we may choose a **sample point** c_k whose value

$$S_P = \sum_{k=1}^n f(c_k) \Delta x_k$$

$f(c_k)$ as the height of the rectangle.

$S_P = \sum_{k=1}^n f(c_k) \Delta x_k$ is called a Riemann Sum of f on $[a, b]$.

Definition The **norm** of a partition P , written $\|P\|$, is the largest of all the subinterval width.

$$\|P\| = \max \{\Delta x_k \mid k=1, \dots, n\}$$

Ex6(p325) The set $P = \{0, 0.2, 0.6, 1, 1.5, 2\}$ is a partition of $[0, 2]$. What is $\|P\|$?

$$\|P\| = 0.5$$

HW5-2

-
- HW: 33,35,39

定積分

5-3 The Definite Integrals

師大工教一

Definition of the Definite Integrals

Definition Let $f(x)$ be a function defined on a close interval $[a,b]$. We say that a number J is the **definite integral of f over $[a,b]$** and that J is the limit of the Riemann sum $\sum_{k=1}^n f(c_k) \Delta x_k$ if the following condition is satisfied:

Given any number $\varepsilon > 0$, there is a corresponding number $\delta > 0$ such that for every partition $P = \{x_0, x_1, x_2, \dots, x_{n-1}, x_n\}$ of $[a, b]$ with $\|P\| < \delta$ and any choice of c_k in $[x_{k-1}, x_k]$, we have $\left| \sum_{k=1}^n f(c_k) \Delta x_k - J \right| < \varepsilon$.

If the limit $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(c_k) \Delta x_k$ exists, we say that the definite integral exists and we write $J = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(c_k) \Delta x_k$.

When the definite integral exists, we say that the Riemann sums of f on

$[a, b]$ converge to the definite integral $J = \int_a^b f(x) dx$ and that f is

integrable over $[a, b]$.

可積分

$$\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(c_k) \Delta x_k \text{ exists,}$$

f is integrable over $[a, b]$

$$\int_a^b f(x) dx = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(c_k) \Delta x_k$$

Consider the case that the subintervals all have the equal width $\Delta x = \frac{b-a}{n}$. If

the definite integral exists, then

$$J = \int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \left(\frac{b-a}{n} \right).$$

P 是 $[a, b]$ n 等分

The Definite Integral as a Limit of Riemann Sums with Equal-Width Subintervals

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f\left(a + k \frac{b-a}{n}\right) \left(\frac{b-a}{n}\right)$$

若樣本為右端點

Ex(102 考古題 #8 加分題) 求 $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \sqrt{1 + \frac{k}{n}} = \lim_{n \rightarrow \infty} \sum_{k=1}^n \sqrt{1 + \frac{k}{n}} \cdot \left(\frac{1}{n}\right)$

(將里曼和(Riemann sum)之極限表成定積分，再求出定積分的值。)

$$P = \left\{1, 1 + \frac{1}{n}, 1 + \frac{2}{n}, \dots, 1 + \frac{n}{n} = 2\right\}, n \text{ 等分}$$

樣本為取右端點

$$R(f, P) = \sum_{k=1}^n f\left(1 + \frac{k}{n}\right) \cdot \frac{1}{n}$$

$$= \sum_{k=1}^n \sqrt{1 + \frac{k}{n}} \cdot \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \sqrt{1 + \frac{k}{n}} \cdot \frac{1}{n} = \int_1^2 \sqrt{x} dx = \frac{4\sqrt{2}-2}{3}$$

Note: $\int_a^b f(x) dx = \int_a^b f(t) dt = \int_a^b f(u) du$: dummy variable.

✗ 只要連續, 就可積分

Theorem 1-Integrability of Continuous Functions

可積分性定理

If a function f is continuous over the interval $[a, b]$, or if f has at most

finitely many jump discontinuities there, then the definite integral $\int_a^b f(x) dx$

exists and f is integrable over $[a, b]$.

Ex1(p328) Show that the function $f(x) = \begin{cases} 1, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$ is not Riemann

integrable over $[0, 1]$.

Let P be any partition, w.l.o.g.,
 $P = \{a = x_0, x_1, \dots, x_n = b\}$.
Let $M_k = \sup\{f(x) \mid x \in [x_{k-1}, x_k]\} = 1$
 $m_k = \inf\{f(x) \mid x \in [x_{k-1}, x_k]\} = 0$.
上和 $U(f, P) = \sum_{k=1}^n M_k \Delta x_k = \sum_{k=1}^n 1 \cdot \Delta x_k = b - a = 1$
下和 $L(f, P) = \sum_{k=1}^n m_k \Delta x_k = 0$.
 $\lim_{\|P\| \rightarrow 0} U(f, P) = 1 \neq \lim_{\|P\| \rightarrow 0} L(f, P) = 0$.
 $\therefore \lim_{\|P\| \rightarrow 0} R(f, P)$ DNE.

Properties of Definite Integrals

Theorem 2 When f, g are integrable over the interval $[a, b]$, the definite integral satisfies the following rules.

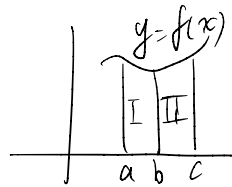
$$1. \int_b^a f(x) dx = -\int_a^b f(x) dx$$

$$2. \int_a^a f(x) dx = 0$$

$$3. \int_a^b k f(x) dx = k \int_a^b f(x) dx$$

$$4. \int_a^b (f(x) \pm g(x)) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$$

$$5. \int_a^c f(x) dx + \int_c^b f(x) dx = \int_a^b f(x) dx$$



$$\int_a^c f(x) dx = \text{area I} + \text{area II}$$

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \text{area I} + \text{area II}$$

6. If f has maximum value $\max f$ and minimum value $\min f$ on $[a, b]$, then

$$(\min f) \cdot (b - a) \leq \int_a^b f(x) dx \leq (\max f) \cdot (b - a)$$

7. If $f(x) \geq g(x)$ on $[a, b]$, then $\int_a^b f(x) dx \geq \int_a^b g(x) dx$.

If $f(x) \geq 0$ on $[a, b]$, then $\int_a^b f(x) dx \geq 0$.

Ex2(p331) Suppose that $\int_{-1}^1 f(x) dx = 5$, $\int_1^4 f(x) dx = -2$, and $\int_{-1}^1 h(x) dx = 7$,

find 1. $\int_4^1 f(x) dx$ 2. $\int_{-1}^1 [2f(x) + 3h(x)] dx$ 3. $\int_{-1}^4 f(x) dx$

$$= 2$$

$$= 31$$

$$= 3$$

Area Under the Graph of a Nonnegative Function

Definition If $y = f(x)$ is nonnegative and integrable over a closed interval

$[a, b]$, then the area under the curve $y = f(x)$ over $[a, b]$ is the integral of

f from a to b , $A = \int_a^b f(x) dx$.

Ex4(p331) Compute $\int_0^b x \, dx$ and find the area A under $y = x$ over the interval $[0, b]$.

$$\int_0^b x \, dx = \frac{1}{2} \times b \times b = \frac{b^2}{2}$$

Extension: 1. $\int_a^b x \, dx = \frac{b^2}{2} - \frac{a^2}{2}.$

2. $\int_a^b c \, dx = c(b - a).$

3. $\int_a^b x^2 \, dx = \frac{b^3}{3} - \frac{a^3}{3}.$

Average Value of a Continuous Function Revisited

Definition If f is integrable on $[a, b]$, then its average value on $[a, b]$, which

is also called its mean, is $av(f) = \frac{1}{b-a} \int_a^b f(x) dx$.

Ex5(p334) Find the average value of $f(x) = \sqrt{4-x^2}$ on $[-2, 2]$.

HW5-3

- HW:3,6,10,16,18,41,49,55,61,65,69

102.1 (4) $\int_0^{2/3} \sqrt{4-9x^2} dx$ → 橢圓 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
Area = πab

$$y = \sqrt{4-9x^2}$$

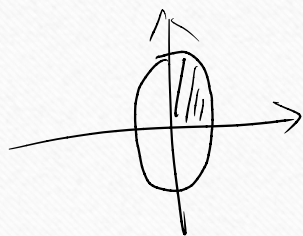
$$y^2 + 9x^2 = 4$$

$$\frac{x^2}{\frac{4}{9}} + \frac{y^2}{4} = 1$$

$$a = \frac{2}{3}$$

$$b = 2$$

$$\frac{1}{4} \int_0^{2/3} \sqrt{4-9x^2} dx = \frac{1}{4} \left(\pi \cdot \frac{2}{3} \cdot 2 \right) = \frac{\pi}{3}$$



5-4 The Fundamental Theorem of Calculus

師大工教一

$$F'(x) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$F'(x) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$F'(x) dx = f(a+dx) - f(a)$$

$$f(x) dx = F(a+dx) - F(a)$$

Theorem 3 The Mean Value Theorem for Definite Integrals

If f is continuous on $[a, b]$, then at some point c in $[a, b]$,

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx$$

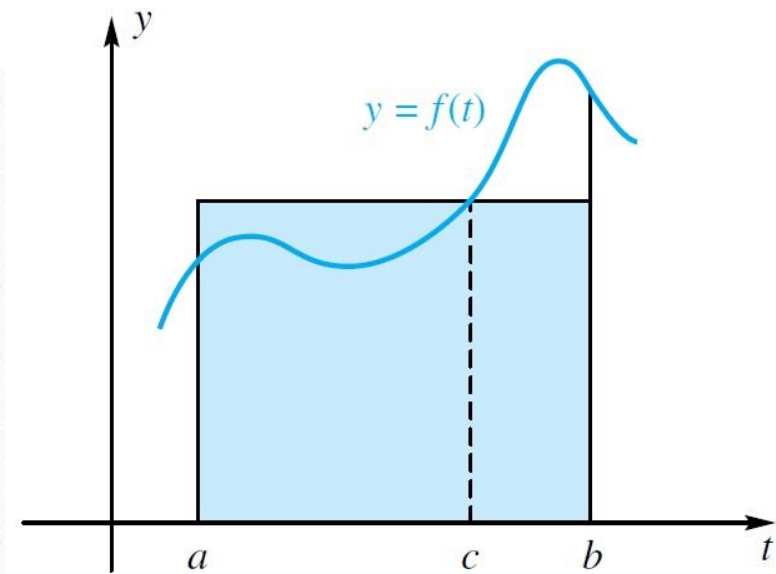


Figure 3

Theorem 4 F.T.O.C. Part I: If f is continuous on $[a, b]$, then

$F(x) = \int_a^x f(t) dt$ is continuous on $[a, b]$ and differentiable on (a, b) , and its

derivative is $f(x)$: $F'(x) = \frac{d}{dx} \int_a^x f(t) dt = f(x)$.

Ex2(p340) Use the Fundamental Theorem to find $\frac{dy}{dx}$ if

(a) $y = \int_a^x (t^3 + 1) dt$ (b) $y = \int_x^5 3t \sin t dt$ (c) $y = \int_1^{x^2} \cos t dt$ (d) $y = \int_{1+3x^2}^4 \frac{1}{2+e^t} dt$

$$\frac{dy}{dx} = x^3 + 1$$

$$\frac{dy}{dx} = -3x \sin x$$

$$\begin{aligned} \frac{dy}{dx} &= \cos(x^2) \cdot 2x - \cos(1) \cdot 0 \\ &= 2x \cos x^2 \end{aligned}$$

$$\frac{dy}{dx} = -\frac{1}{2+e^{1+3x^2}} \cdot 6x$$

$$\begin{aligned} F(x) &= \int_a^x f(t) dt. \text{ We want to find } \\ \frac{d}{dx} \left[\int_{h(x)}^{g(x)} f(t) dt \right] &= \int_a^{g(x)} f(t) dt - \int_a^{h(x)} f(t) dt \\ &= F(g(x)) - F(h(x)) \\ \frac{d}{dx} \left[\int_{h(x)}^{g(x)} f(t) dt \right] &= F'(g(x)) \cdot g'(x) - F'(h(x)) \cdot h'(x) \\ &= f(g(x)) \cdot g'(x) - f(h(x)) \cdot h'(x) \end{aligned}$$

Ex(#2, 105 年考古題) 已知 $\int_0^{x^5} f(2t^2 - t + 1) dt = \ln x$ ，求 $f(2017)$ 之值。

$$f(2x^{10} - x^5 + 1) \cdot (5x^4) = \frac{1}{x}$$

$$x = 2017 \quad f(2 \cdot 2017^{10} - 2017^5 + 1)(5 \cdot 2017^4) = \frac{1}{2017}$$

$$f(2017) \cdot 80 = \frac{1}{2}$$

$$f(2017) = \frac{1}{160}$$

Theorem 4 F.T.O.C. Part II: If f is continuous over $[a, b]$, and F is any antiderivative of f on $[a, b]$, then $\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a)$.

Ex3(p342) Find the following definite integrals:

$$(a) \int_0^{\pi} \cos x \, dx$$

$$(b) \int_{-\frac{\pi}{4}}^0 \sec x \tan x \, dx$$

$$(c) \int_1^4 \left(\frac{3}{2} \sqrt{x} - \frac{4}{x^2} \right) dx$$

$$(d) \int_0^1 \frac{1}{x+1} \, dx$$

$$(e) \int_0^1 \frac{1}{x^2 + 1} \, dx = \arctan x \Big|_0^1 = \arctan(1) - \arctan(0) = \frac{\pi}{4} - 0 = \frac{\pi}{4}$$

$$\begin{aligned} (a) \int_0^{\pi} \cos x \, dx &= \sin x \Big|_0^{\pi} \\ &= 0 \end{aligned}$$

$$\begin{aligned} (b) &= \sec x \Big|_{-\frac{\pi}{4}}^0 \\ &= 1 - \frac{1}{\cos(-\frac{\pi}{4})} = 1 - \sqrt{2} \end{aligned}$$

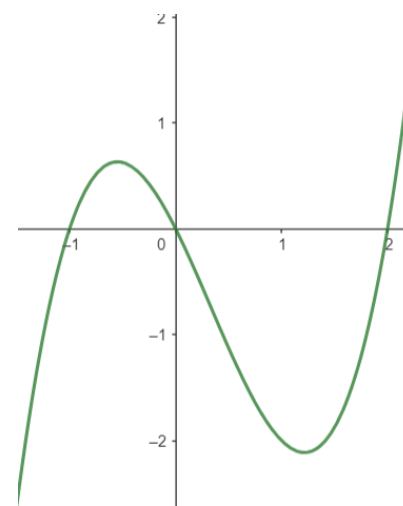
$$\begin{aligned} (c) &= x^{\frac{3}{2}} + 4x^{-1} \Big|_1^4 \\ &= 8 + 1 - 1 - 4 = 4 \end{aligned}$$

$$\begin{aligned} (d) &= \ln|x+1| \Big|_0^1 \\ &= \ln 2 - \ln 1 = \ln 2 \end{aligned}$$

Total Area

To find the area between the graph of $y = f(x)$ and the x -axis over the interval $[a, b]$:

1. Subdivide $[a, b]$ at the zeros of f .
2. Integrate f over each subintervals.
3. Add the absolute values of the integrals.



Ex8(p346) Find the area of the region and the x -axis and the graph of

$$f(x) = x^3 - x^2 - 2x, -1 \leq x \leq 2.$$

$$\begin{aligned} \text{Total area} &= \int_{-1}^0 (x^3 - x^2 - 2x) dx - \int_0^2 (x^3 - x^2 - 2x) dx \\ &= \left(\frac{1}{4}x^4 - \frac{1}{3}x^3 - x^2 \right) \Big|_{-1}^0 - \left(\frac{1}{4}x^4 - \frac{1}{3}x^3 - x^2 \right) \Big|_0^2 \\ &= \left(\frac{1}{4} - \frac{1}{3} - 1 \right) - \left(\frac{16}{4} - \frac{8}{3} - 4 \right) \\ &= -\frac{1}{4} - \frac{1}{3} + 1 + \frac{8}{3} - 4 + 4 \\ &= -\frac{1}{4} - \frac{1}{3} + 1 + \frac{8}{3} = -\frac{1}{4} + \frac{7}{3} + 1 = \frac{37}{12} \end{aligned}$$

HW5-4

- HW:1,6,7,10,17,22,29,39,43,47,48,83.

5-5 Indefinite Integrals and the Substitution Method

代換法

師大工教一

Recall: In Section 4.8, we have defined the indefinite integral

$$\int f(x) dx = F(x) + C, \text{ where } F \text{ is any antiderivative of } f.$$

Substitution: Running the Chain Rule Backwards

$$\text{E.g. } \frac{d}{dx} \left(\frac{1}{2} \ln |2 \sec x - 1| \right) = \frac{\sec x \tan x}{2 \sec x - 1} \Rightarrow \int \frac{\sec x \tan x}{2 \sec x - 1} dx = \frac{1}{2} \ln |2 \sec x - 1| + C$$

Ex1(p350) Find the integral $\int (x^3 + x)^5 (3x^2 + 1) dx$.

$$\begin{aligned}\text{Let } u &= x^3 + x \\ du &= (3x^2 + 1) dx\end{aligned}$$

$$\begin{aligned}&= \int u^5 du \\&= \frac{u^6}{6} + C \\&= \frac{(x^3 + x)^6}{6} + C\end{aligned}$$

$$[f(g(x))]' = f'(g(x)) \cdot g'(x)$$

$$\Rightarrow \int f'(g(x)) \cdot g'(x) = f(g(x)) + C$$

Theorem 6-The Substitution Rule

If $u = g(x)$ is a differentiable function whose range is in an interval I , and

f is continuous on I , then $\int f(g(x)) \cdot g'(x) dx = \int f(u) du$.

The Substitution Method to evaluate $\int f(g(x)) \cdot g'(x) dx$

1. Substitute $u = g(x)$ and $du = \left(\frac{du}{dx}\right)dx = g'(x)dx$ to obtain $\int f(u)du$.

2. Integrate with respect to u .

3. Replace u by $g(x)$.

Ex4(p352) Find $\int \cos(7\theta + 3) d\theta$.

$$\text{Let } u = 7\theta + 3$$

$$du = 7 d\theta$$

$$d\theta = \frac{du}{7}$$

$$\text{origin} = \int \cos(u) \frac{du}{7} = \frac{1}{7} \sin u + C = \frac{1}{7} \sin(7\theta + 3) + C$$

Ex(105#1(4)) Find $\int \frac{\sec x \tan x}{2 \sec x - 1} dx$.

$$\text{Let } u = 2 \sec x - 1$$

$$du = 2 \sec x \tan x \, dx$$

$$\frac{du}{2} = \sec x \tan x \, dx$$

$$\begin{aligned} \int \frac{\frac{du}{2}}{u} &= \frac{1}{2} \int \frac{1}{u} \, du = \frac{1}{2} \ln|u| + C \\ &= \frac{1}{2} \ln|2 \sec x - 1| + C \end{aligned}$$

Ex6(p353) Evaluate $\int x\sqrt{2x+1} dx$.

$$\text{Let } u = 2x + 1$$

$$du = 2 dx$$

$$dx = \frac{du}{2}$$

$$x = \frac{u-1}{2}$$

$$\text{origin} = \int \frac{u-1}{2} \sqrt{u} \frac{du}{2}$$

$$= \frac{1}{4} \int (u-1) \sqrt{u} du$$

$$= \frac{1}{4} \int u\sqrt{u} - \sqrt{u} du$$

$$= \frac{1}{4} \int u^{\frac{3}{2}} - u^{\frac{1}{2}} du$$

$$= \frac{1}{4} \left(\frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} \right) + C$$

$$= \frac{u^{\frac{5}{2}}}{10} - \frac{u^{\frac{3}{2}}}{6} + C$$

$$= \frac{(2x+1)^{\frac{5}{2}}}{10} - \frac{(2x+1)^{\frac{3}{2}}}{6} + C$$

Ex7(p354) Evaluate the following integrals:

(a) $\int \sin^2 x \, dx$ (b) $\int \cos^2 x \, dx$ (c) $\int \tan x \, dx$

$$(a) = \int \frac{1 - \cos 2x}{2} \, dx$$

$$= \frac{1}{2} \int (1 - \cos 2x) \, dx$$

$$= \frac{1}{2} \left(x - \frac{1}{2} \sin 2x \right) + C$$

$$(b) = \frac{1}{2} \int (1 + \cos 2x) \, dx$$

$$= \frac{1}{2} \left(x + \frac{1}{2} \sin 2x \right) + C$$

$$(c) = \int \frac{\sin x}{\cos x} \, dx$$

Let $u = \cos x$
 $du = -\sin x \, dx$

$$\begin{aligned} &= \int \frac{-du}{u} = -\ln|u| + C \\ &= -\ln|\cos x| + C \\ &= \ln|\sec x| + C \end{aligned}$$

倍角公式
 $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$
 $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$

$$\left\{ \begin{array}{l} \sin^2 \theta + \cos^2 \theta = 1 \\ \cos^2 \theta = 1 - \sin^2 \theta \\ \cos 2\theta = \cos^2 \theta - \sin^2 \theta \\ \sin^2 \theta = \cos^2 \theta - \cos 2\theta \\ \quad = 1 - \sin^2 \theta - \cos 2\theta \\ 2\sin^2 \theta = 1 - \cos 2\theta \\ \sin^2 \theta = \frac{1 - \cos 2\theta}{2} \end{array} \right.$$

Ex8(p354) Evaluate the following integrals:

(a) $\int \frac{dx}{e^x + e^{-x}}$ (b) $\int \sec x \, dx$

$$(a) = \int \frac{dx}{e^x + e^{-x}} \cdot \frac{e^x}{e^x}$$

$$= \int \frac{e^x dx}{e^{2x} + 1}$$

$$\text{Let } u = e^x \\ du = e^x dx$$

$$= \int \frac{du}{u^2 + 1} = \tan^{-1} u + C \\ = \tan^{-1} e^x + C$$

$$(b) = \int \sec x dx \cdot \frac{\sec x + \tan x}{\sec x + \tan x}$$

$$\text{Let } u = \sec x + \tan x$$

$$du = (\sec x \tan x + \sec^2 x) dx$$

$$= \sec x (\sec x + \tan x) dx$$

$$= \int \frac{du}{u} = \ln |u| + C \\ = \ln |\sec x + \tan x| + C$$

Formulas

$$\int \tan x \, dx = \ln |\sec x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C$$

$$\int \cot x \, dx = -\ln |\csc x| + C$$

$$\int \csc x \, dx = -\ln |\csc x + \cot x| + C$$

HW5-5

- HW: 2,3,4,11,16,24,38,55,57,67

5-6 Definite Integral Substitutions and the Area Between Curves

師大工教一

Theorem 7-Substitution in Definite Integrals

If g' is continuous on the interval $[a, b]$ and f is continuous on the range

of $g(x) = u$, then $\int_a^b f(g(x)) \cdot g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$.

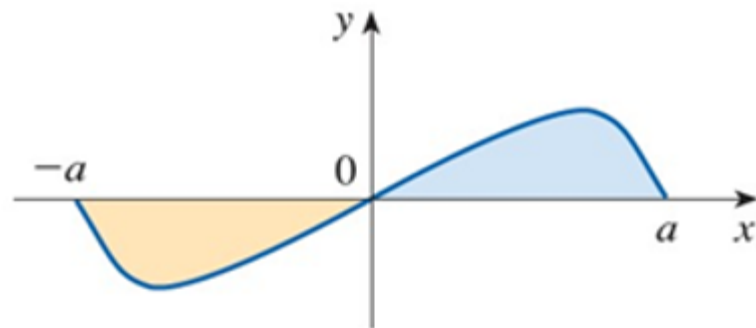
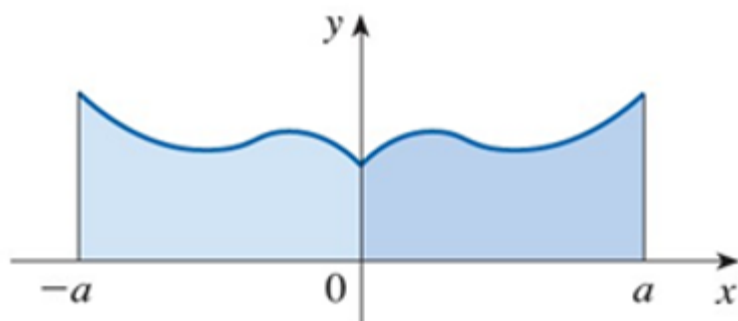
Ex2(p359) Evaluate the following definite integrals:

(a) $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot \theta \csc^2 \theta d\theta$ (b) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \tan \theta d\theta$

Let $u = \cot \theta$
 $du = -\csc^2 \theta d\theta$
 $-du = \csc^2 \theta d\theta$
origin $= \int_1^0 u (-du)$
 $= -\int_1^0 u du = \int_0^1 u du$
 $= \frac{u^2}{2} \Big|_0^1 = \frac{1}{2}$

$= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sin \theta}{\cos \theta} d\theta$
Let $u = \cos \theta$
 $du = -\sin \theta d\theta$
 $-du = \sin \theta d\theta$
 $= \int_{\frac{1}{\sqrt{2}}}^1 \frac{-du}{u} = 0$

Definite Integrals of Symmetric Functions



Theorem 8 Let f be continuous on the symmetric interval $[-a, a]$.

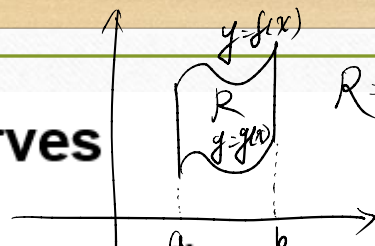
(a) If f is even, then $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$.

(b) If f is odd, then $\int_{-a}^a f(x) dx = 0$.

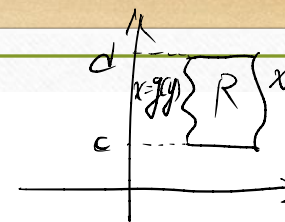
Ex3(p360) Evaluate $\int_{-2}^2 (x^4 - 4x^2 + 6) dx$.



Area Between Curves



$$R = \int_a^b (f(x) - g(x)) dx$$



$$R = \int_c^d (f(y) - g(y)) dy$$

Definition If f, g are continuous with $f(x) \geq g(x)$ throughout $[a, b]$, then

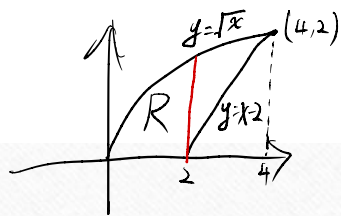
the **area of the region between the curve $y = f(x)$ and $y = g(x)$ from a**

to b is the integral of $(f - g)$ from a to b : $A = \int_a^b (f(x) - g(x)) dx$.

Ex6(p362) Find the area of the region in the first quadrant that is bounded

above by $y = \sqrt{x}$ and below by the x -axis and the line $y = x - 2$. (Figure

5.30)



$$\int_0^2 \sqrt{x} dx + \int_2^4 [\sqrt{x} - (x-2)] dx$$

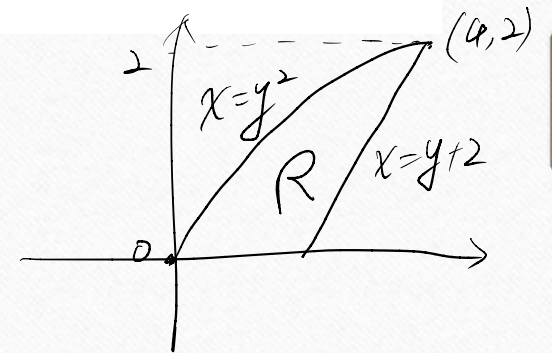
$$= \left[\frac{2}{3} x^{\frac{3}{2}} \right]_0^2 + \left[\frac{2}{3} x^{\frac{3}{2}} - \frac{x^2}{2} + 2x \right]_2^4$$

$$= \frac{4\sqrt{2}}{3} + \left[\left(\frac{16}{3} - 8 + 8 \right) - \left(\frac{4\sqrt{2}}{3} - 2 + 4 \right) \right] = \frac{16}{3} - 2 = \frac{10}{3}$$

Integration with Respect to y

Ex7(p363) Find the area of the region in Exmple 6 by integrating with respect to y . (region in the first quadrant that is bounded above by $y = \sqrt{x}$ and below by the x -axis and the line $y = x - 2$).

$$\begin{aligned}\text{Area} &= \int_0^2 [(y+2) - (y^2)] dy \\ &= \left. \frac{1}{2}y^2 + 2y - \frac{1}{3}y^3 \right|_0^2 \\ &= \left(2 + 4 - \frac{8}{3} \right) = \frac{10}{3}\end{aligned}$$



HW5-6

- HW: 2,5,9,13,29,39,48,51,55,60,82,85,106